

Edge-Oriented Multi-Task Vision for Shrimp Detection and Residual-Feed Segmentation

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Abstract

Routine shrimp-farm management relies on observations of shrimp presence and growth together with assessments of residual feed. These quantities are conventionally obtained through periodic sampling and manual inspection, which require operator labor, may involve animal handling, and provide intermittent snapshots of pond conditions. Underwater cameras combined with deep vision models offer a non-contact means of collecting observations more frequently by detecting visible shrimp, identifying detections suitable for downstream length measurement, and segmenting residual feed. Continuous cloud inference requires sustained video transmission and reliable pond-side connectivity. Local inference reduces this dependency, but independently executing detection and segmentation networks duplicates feature extraction and restricts throughput on resource-constrained edge devices.

We present a unified PIDNet-YOLO network that produces residual-feed segmentation and two-state shrimp detections from one shared PIDNet traversal. The detector distinguishes clean shrimp suitable for downstream length measurement from count shrimp that remain useful for monitoring visible presence. We remove the GELAN backbone of YOLOv9-T and connect scale-aligned proportional, integral, and derivative features from PIDNet to the retained YOLO neck and detection head through learned spatial resamplers and depthwise-pointwise adapters. Because feed masks and shrimp boxes belong to separate annotation pools, we use a three-stage curriculum comprising shrimp-oriented representation learning, protected feed adaptation, and detector recovery against the completed frozen PIDNet. Boundary-Privileged Knowledge Distillation supports feed adaptation in a PIDNet-S-0.75 variant with reduced channel capacity.

On the designated session-disjoint holdout, treated as a secondary non-blind estimate, compact fusion ob-

tains $AP_{50:95}/mIoU$ values of 0.5442/0.7721 at 448×448 and 0.5805/0.7427 at 640×640 , compared with 0.5834/0.7591 and 0.6032/0.7684 for width-matched separate models. The compact fused graph reduces parameter count by 9.3% and computation by 14.2%. On a four-core Raspberry Pi 5 CPU, fusion increases end-to-end throughput from 13.525 to 16.401 FPS at 448 and from 6.015 to 7.328 FPS at 640, while reducing peak memory by 21.87 and 34.41 MiB, respectively. In a controlled cohort, the fused configurations achieve conditional length MAPE of 6.05–6.18% while covering 90.67–92.67% of 150 shrimp; these errors apply to selected, annotation-matched detections. Analysis of 258,809 frames from three historical days additionally identifies 27 model-consensus visible-feed period candidates spanning 95 minutes. These candidates require visual confirmation because the archive contains no feed-event annotations. The results characterize the predictive and operational trade-offs of shared multi-task inference for local shrimp-farm monitoring.

1. Introduction

Feed can account for 40–65% of shrimp-production costs [21], while uneaten feed contributes organic matter and nutrients that degrade pond conditions [5]. Farmers therefore inspect feed trays and periodically capture shrimp to assess feeding and growth. These procedures require recurring labor, provide only intermittent observations, and, for length measurement, require animal handling [2, 24]. Underwater cameras and computer vision offer a non-contact alternative that can observe residual feed, visible shrimp, and measurement-eligible shrimp more frequently.

Prior work has demonstrated shrimp counting in farm environments [1], distinguished measurable from merely visible shrimp for downstream length estimation [11], and detected residual feed either independently or jointly with shrimp [7, 8, 12]. Continuous

075 monitoring, however, also requires efficient deployment.
076 Cloud inference depends on network availability and
077 image-transfer bandwidth, which may be limited at
078 aquaculture sites [18]. Edge inference reduces this
079 dependence [19], but embedded devices have substan-
080 tially smaller compute and memory budgets. Running
081 independent detection and segmentation networks on
082 every frame duplicates feature extraction and increases
083 latency, memory, and energy demand.

084 A shared encoder can avoid this duplication, as de-
085 tection and semantic segmentation can be produced
086 from common features in one forward pass [3]. Two is-
087 sues complicate sharing in our setting. First, residual-
088 feed segmentation requires spatial and boundary de-
089 tail, whereas shrimp detection requires scale-aligned
090 features for classification and localization. Second, feed
091 masks and shrimp boxes belong to separate annota-
092 tion pools, so most images supervise only one task. Al-
093 though missing-task losses can be omitted during het-
094 erogeneous multi-task training [10], alternating objec-
095 tives can still overwrite features needed by the other
096 task.

097 We develop a unified PIDNet-YOLO architec-
098 ture for simultaneous residual-feed segmentation and
099 shrimp detection on pond-side hardware. PIDNet-S
100 supplies the shared detail, context, and boundary rep-
101 resentations and retains its native binary segmenta-
102 tion route [25]. Lightweight resamplers and depthwise-
103 pointwise adapters map its multiscale features to the
104 retained YOLOv9-T neck and detection head, replac-
105 ing the detector’s GELAN backbone [22]. The detec-
106 tor separates shrimp suitable for an external length-
107 measurement procedure (clean) from other detectable
108 shrimp used for visible-presence monitoring (count); it
109 predicts measurement eligibility, not physical length.

110 We accommodate disjoint annotations with a
111 three-stage curriculum. Detection training first ex-
112 poses the shared representation to shrimp localization
113 cues. Feed-segmentation adaptation then keeps the
114 detection-oriented derivative branch active but freezes
115 its parameters and statistics. Finally, detector-side
116 components are trained against the feed-adapted en-
117 coder while the segmentation route is preserved. For
118 the reduced-width PIDNet-S-0.75 model, Boundary-
119 Privileged Knowledge Distillation transfers segmen-
120 tation knowledge from a full-width teacher without
121 adding a deployment branch [16]. We evaluate sharing,
122 width reduction, and input resolution using matched
123 predictive metrics, a designated session-disjoint hold-
124 out, complexity analysis, and measured four-core Rasp-
125 berry Pi 5 CPU inference at 448 and 640 pixels. We
126 additionally assess controlled length measurements and
127 three days of historical video.

Our contributions are:

- A shared-encoder network that produces residual-
feed masks and clean/count shrimp detections in
one forward pass, avoiding a second image encoder
through lightweight multiscale PIDNet-YOLO in-
terfaces.
- A three-stage curriculum for disjoint segmentation
and detection annotations, with boundary-aware
distillation supporting native 0.75-width scaling
without inference-time overhead.
- Matched accuracy-efficiency evaluation at two res-
olutions, including Raspberry Pi 5 CPU measure-
ments; the compact fused model reaches approx-
imately 1.21× the throughput of sequential task-
specific inference at both resolutions while reducing
latency and peak memory.
- Downstream evaluation on controlled-length data
and historical video, demonstrating length analysis
and identification of candidate visible-feed periods
from the fused outputs.

2. Related Work

2.1. Shrimp and Feed Monitoring

Computer vision has supported shrimp counting in con-
trolled and farm environments [1, 26]. Lai et al. sepa-
rated measurable shrimp from animals visible but un-
suitable for length estimation [11]; this distinction mo-
tivates our clean and count observation classes. Feed-
related studies have jointly detected shrimp and left-
over feed, segmented residual feed, or combined shrimp
tracking with residue quantification [7–9, 12]. Multi-
indicator systems have also combined counting, size,
feeding, or weight analyses through task-specific net-
works and downstream modules [13, 23]. These sys-
tems establish the required visual capabilities, but mod-
ular pipelines can process the same frame repeatedly.
We instead target feature reuse between pixel-level feed
segmentation and shrimp detection.

2.2. Shared Perception with Disjoint Annotations

Shared multi-task networks reduce duplicated compu-
tation by coupling a common encoder to task-specific
outputs. MultiNet jointly addresses scene classifica-
tion, vehicle detection, and road segmentation, while
BlitzNet predicts detection and semantic segmenta-
tion from shared multiscale features [3, 20]. FPN and
PANet further demonstrate the importance of mul-
tiscale top-down and bottom-up feature aggregation
[15, 17]. Our architecture similarly reuses features,
but connects PIDNet’s detail, context, and boundary
streams to a retained YOLO neck.

Disjoint annotations complicate optimization. Uber-

Net sets an unannotated task’s loss to zero for that sample [10], avoiding unsupported supervision but not preventing updates for one task from changing features used by another. This is especially relevant here because PIDNet’s dense segmentation representation and YOLOv9’s P3–P5 detection features have different spatial requirements [22, 25]. Our staged curriculum protects task-relevant representations while retaining one deployment graph.

Reducing encoder width adds a capacity constraint. Knowledge distillation can transfer behavior from a larger teacher without retaining it at inference [6]. Boundary-Privileged Knowledge Distillation separates classwise interior and boundary information for semantic segmentation [16]; we use it only during reduced-width feed adaptation.

2.3. Aquaculture Edge Deployment

Edge inference processes imagery near the camera but shifts computation to devices with tighter memory and processing limits. Aquaculture platforms have investigated embedded nodes for local inference and workload coordination [4]. Prior multi-indicator systems primarily retain task-specific models or processing stages, and therefore do not provide a matched hardware comparison of independent versus shared detection–segmentation inference. We fill this gap by comparing sequential baselines and the fused graph under the same runtime conditions on a four-core Raspberry Pi 5 CPU, reporting accuracy, model complexity, latency, throughput, and peak memory at two resolutions.

3. Method

The model produces a binary residual-feed mask and two-class shrimp detections in one pass. It combines a shared PIDNet–YOLO graph, a three-stage curriculum for disjoint annotations, and a reduced-width, distilled edge configuration. All teachers and auxiliary paths are removed for deployment.

3.1. PIDNet–YOLO Fusion

PIDNet separates spatial detail, context, and boundaries into proportional (P), integral (I), and derivative (D) paths, respectively; PagFM exchanges path information, PAPPm aggregates context, and the D-guided final fusion (DFM, implemented by Light_Bag) forms the semantic representation [25]. We preserve this native PIDNet-S segmentation route and remove YOLOv9-T layers 0–8 (its GELAN backbone), retaining its multiscale neck, anchor-free heads, and training-only Programmable Gradient Information (PGI) path [22].

Table 1. Detector-interface sources in concatenation order.

Input	PIDNet sources	Strides	Output
A_3	$p_{l3}, stem, d_{l3}$	8, 8, 8	64, $s=8$
A_4	p_{l4}, i_{l3}, d_{l4}	8, 16, 8	96, $s=16$
A_5	p_{l5}, i_{l4}, d_{l5}	8, 32, 8	128, $s=32$

Detection-privileged D path. Because the feed task has one foreground class, we reserve D’s boundary capacity principally for detection. D enters all detector interfaces and remains forward-active in Light_Bag, but its parameters and normalization state are protected during feed adaptation.

Multiscale interfaces. Table 1 gives the sources for the stride-8/16/32 neck inputs. Here p_{l3} and p_{l4} are tapped immediately before their PagFM stages; p_{l5} supplies the deepest interface because PIDNet has no third PagFM. The A_3 middle source is the stride-8 shared stem before the P/I split.

P and D stay at stride 8. Each A_4 P/D source is aligned by an independent depthwise 3×3 , stride-2 convolution with BN and SiLU; each A_5 source uses an independent two-stage cascade of the same operation, not the A_4 tensors. A_3 needs no reduction. With $C_3 = stem$, $C_4 = i_{l3}$, and $C_5 = i_{l4}$, the adapters are

$$Z_k = [\tilde{P}_k \| C_k \| \tilde{D}_k], \quad A_k = \phi_k^{pw}(\phi_k^{dw}(Z_k)), \quad (1)$$

for $k \in \{3, 4, 5\}$. The stride-1 depthwise 3×3 operation refines each channel; the pointwise 1×1 operation mixes sources and projects to 64/96/128 channels. Both use BN and SiLU. The retained neck begins with SPPELAN at A_5 and preserves its top-down/bottom-up aggregation. PGI consumes the same A_3 – A_5 tensors during training and is structurally removed with its auxiliary towers for inference.

3.2. Three-Stage Training for Disjoint Supervision

Feed masks and shrimp boxes belong to separate, unequal annotation pools. We first make the shared representation detection-aware, then adapt its segmentation components to feed, and finally recover detection against the completed encoder. An image contributes only its available objective.

Stage 1: detection-oriented representation learning. Shrimp-box losses update every PIDNet component on a route to the selected P/context/D taps, the learned resamplers and adapters, and the retained neck, PGI path, and detection towers. Semantic-only PIDNet components receive no detection gradient; no feed

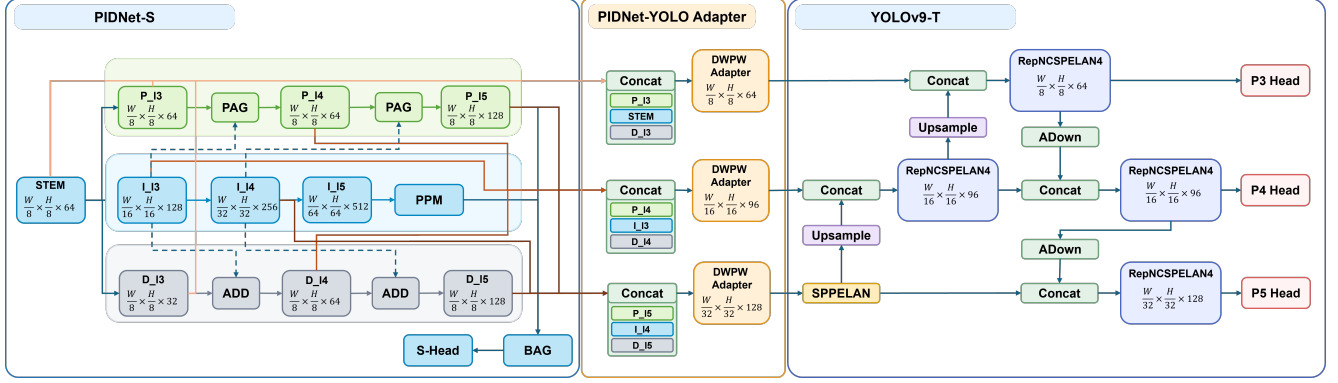


Figure 1. Full-width fused deployment graph. PIDNet-S retains its semantic output and supplies aligned P/context/D features through depthwise-pointwise adapters to the YOLOv9-T neck and main head. PAG, PPM, and BAG denote PagFM, PAPPm, and DFM; training-only heads and PGI are omitted.

loss or Stage-2 auxiliary semantic/boundary head is present.

Stage 2: protected feed adaptation. The full Stage-1 PIDNet initializes a detector-free feed graph. Feed losses update the stem through P/I routes, PagFM, PAPPm, Light_Bag, and the semantic heads. D remains forward-active but isolated: its parameters are frozen, its modules and BN stay in evaluation mode, shared inputs entering D are detached, and D outputs entering Light_Bag and the auxiliary boundary head are detached. Thus feed gradients reach the stem through P/I, not D. D activations may change with the updated stem; the protected objects are D’s learned function and normalization state. The supervised loss is

$$\mathcal{L}_{\text{seg}} = 0.4\mathcal{L}_{\text{aux}}^{\text{WCE}} + \mathcal{L}_{\text{main}}^{\text{OHem}} + 20\mathcal{L}_{\text{bdry}}^{\text{DBCE}} + \mathcal{L}_{\text{sem-bdry}}^{\text{OHem}}, \quad (2)$$

where WCE, OHem, and DBCE are weighted cross-entropy, online hard-example mining, and dynamically balanced binary cross-entropy. Auxiliary heads are discarded after this stage.

Stage 3: detector recovery. We overlay the Stage-2 PIDNet state on the Stage-1 fused graph and restore the Stage-1 detector state. The complete PIDNet and all PIDNet BN statistics are frozen in evaluation mode. Native shrimp-box losses update only resamplers, adapters, the main neck, PGI, and both detection towers; no detection teacher is used. Removing PGI leaves the Stage-2 segmenter and recovered main detector.

3.3. Edge-Oriented Optimization

The compact model instantiates PIDNet-S at 0.75× width without pruning; the main/PAPPm/head con-

trols change from 32/96/128 to 24/72/96, with derived widths following them. Compatible ImageNet weights are initialized by deterministic channel-prefix slicing. Adapter inputs become 120/192/384 channels, while their 64/96/128 outputs and YOLO topology remain unchanged. The 448 and 640 models are trained independently.

During Stage 2, Boundary-Privileged Knowledge Distillation (BPKD-v3) transfers the final two-class, stride-8 semantic logits from a frozen, validation-selected PIDNet-S teacher trained on the same feed split [16]. Teacher and student share the resolution and augmented image. Classwise trimaps define interior and transition regions for body and edge transfer:

$$\mathcal{L}_{S2} = \mathcal{L}_{\text{seg}} + \lambda_e (20\mathcal{L}_{\text{body}}^{\text{KD}} + 50\mathcal{L}_{\text{edge}}^{\text{KD}}), \quad \lambda_e = \min(0.02e, 0.10). \quad (3)$$

Temperature is 1; edge weights are 2 for feed and 1 for background. D protection remains active, and KD gradients enter only through the final semantic-logit route. The compact model uses a full-width teacher; the full-width student uses a same-width teacher. Teachers, trimaps, auxiliary heads, PGI, and auxiliary detection towers are absent at deployment.

4. Experimental Setup

Validation data select checkpoints and models. The designated session-disjoint holdout is reported separately for the finalized inventory; because it was consulted during legacy development, it is secondary evidence, not a blind confirmatory test. Application thresholds were fixed before the deployment and historical-video analyses.

Table 2. Perception dataset after grouping.

Split	Det. images	Boxes	Seg. images
Train	7,121	26,008	1,619
Validation	898	2,945	206
Holdout	967	3,560	200

4.1. Datasets and Annotations

Detection and segmentation use disjoint annotation pools. Shrimp boxes are clean when visibility and pose support downstream length measurement, or count when useful only for visible-presence monitoring; these labels describe observational utility, not animal condition. Segmentation masks mark residual feed versus background. Missing annotations are never treated as negative labels for the other task.

Before the 80/10/10 split, images from multiple cameras were grouped jointly by capture session and exact decoded-image identity; conflicting duplicates were removed. The partitions are session-disjoint, but not farm-disjoint.

A separate controlled-length cohort contains 150 physical shrimp in 1,050 annotated views. Its historical split manifest is unavailable, so it is an application cohort rather than an independent perception test. All imagery was previously collected at a private shrimp farm in Taiwan and used with permission; source mappings and full datasets are withheld by agreement.

4.2. Training Configuration

The component states and losses for the three stages follow Section 3.2; PGI and auxiliary heads are removed for evaluation. Table 3 summarizes optimization.

Detection augmentation combines photometric and geometric transforms with mosaic, mixup, and copy-paste; mosaic closes for the last 15 epochs. Segmentation uses 0.5–2.1 random scaling, square crop/pad, and horizontal flip. The frozen, validation-selected Stage-2 PIDNet-S teacher uses the same feed split, augmented image, and resolution as its student. BPKD trimap kernels are 7×7 at 448 and 9×9 at 640. Further augmentation, OHEM, and distillation settings are supplied in the supplement.

The 448 and 640 models are trained independently. Compact 448 repeats share Stages 1–2 and repeat Stage 3; compact 640 repeats train all three stages independently. Full-width models and most ablations use seed 0. Runs use one 32-GB NVIDIA RTX 5090, PyTorch 2.12.1, and CUDA 13.2.

4.3. Metrics

Detection reports class-averaged precision and recall at maximum smoothed mean F1, AP_{50} , COCO $AP_{50:95}$,

and classwise $AP_{50:95}$ [14]. Evaluation uses the PGI-removed FP32 graph, centered square letterboxing, confidence floor 0.001, NMS IoU 0.70, and no test-time augmentation. Segmentation logits are bilinearly resized with `align_corners=true`; one aggregate confusion matrix yields mean IoU, feed IoU, and feed Dice $2TP/(2TP + FP + FN)$.

Parameters and GFLOPs describe the batch-one deployment graph after removing training-only components; GFLOPs are twice THOP’s multiply-accumulate count. Separate-system values sum standalone PIDNet and YOLO. Raspberry Pi reporting includes preprocessing, inference, postprocessing, total compute, end-to-end FPS including decoding, and peak resident set size (RSS).

Controlled-length evaluation reports MAE, MAPE, valid-view coverage, and physical-shrimp coverage after averaging valid views per shrimp; 95% intervals use 5,000 shrimp-level bootstrap replicates for fixed checkpoints. Historical analysis reports a minute-balanced visible-count index and model-consensus visible-feed period candidates, not unique-animal counts or feed mass/consumption. Historical length is not included in the main analysis.

4.4. Deployment and Application Protocols

Raspberry Pi 5. NCNN mixed-FP16 graphs run without an accelerator on an actively cooled Raspberry Pi 5 Model B Rev. 1.0, using four threads pinned to its four CPU cores. At each resolution, fused PIDNet-S-0.75 is compared with sequential standalone PIDNet-S-0.75 and YOLOv9-T on the same video. Detection uses confidence 0.25 and NMS IoU 0.45. Three interleaved repetitions each contain 20 warm-up and 600 measured frames; a 45 °C cooldown gate precedes each run, and throttling flags must remain zero. Display and encoding are disabled.

Controlled length. For each view, the downstream procedure selects the largest clean detection with confidence ≥ 0.10 ; it is valid when box IoU with the annotation is ≥ 0.50 . The fixed Report-GL3 diagonal calibration converts the box to length. Valid views are averaged per shrimp before error calculation. Thus reported error is conditional on successful selection and matching; the fused network predicts eligibility and a box, not length itself.

Historical video. The archive contains 258,809 common frames sampled at 1 FPS over three predeclared days (approximately 71.9 hours). We compare separate full-width PIDNet-S+YOLOv9-T and fused PIDNet-S-0.75, each at 448 and 640; this is an operational deploy-

Table 3. Three-stage training configuration.

Setting	Stages 1 and 3	Stage 2
Optimizer	SGD, Nesterov momentum 0.937	Adam, $\beta = (0.5, 0.999)$
Initial LR	0.01	0.001, polynomial decay
Batch	32; nominal 64 by accumulation	32, no accumulation
Weight decay	0.0005	0.0005
Epoch limit / patience	200 / 50	200 / 50
Selection	$0.1AP_{50} + 0.9AP_{50:95}$	mIoU

ment comparison, not a width-matched control. Class-aware NMS uses IoU 0.45, and both shrimp classes contribute to visible count at confidence ≥ 0.60 . Observations are first averaged within each minute, then valid minutes receive equal weight.

A system is feed-positive in a local minute when its 31-second rolling-median feed area reaches 1.0% of image area. A minute is consensus-active when at least three of four systems are positive. Periods bridge at most one inactive or missing minute and require at least two active minutes; no local-floor subtraction is used. With no frame-level feed labels or event annotations, these are only sustained feed-like temporal candidates. Exact windows and per-system statistics appear in the supplement.

5. Results

Perception is reported on validation and a designated session-disjoint holdout (967 detection images, 3,560 boxes, and 200 segmentation images). Because the holdout was accessed during legacy development, it is a secondary, non-blind estimate; all ablations use validation only.

5.1. Perception and Edge Efficiency

Table 4 compares separate PIDNet+YOLO execution with one shared PIDNet traversal. Compact 448 detection is the mean of three Stage-3 recoveries from one Stage-2 state; compact 640 repeats the complete curriculum three times. Run-level metrics and dispersion are supplementary.

At compact 448, fusion raises feed IoU by 2.58 validation points and 2.43 holdout points, while $AP_{50:95}$ falls by 1.59 and 3.92 points. At compact 640, the AP gaps are 2.15 and 2.27 points; holdout feed IoU is 4.84 points lower. Full-width fusion narrows the holdout AP gap to 1.14 points (0.52 at AP_{50}) while recall rises 0.39 point. Its validation mIoU is 0.49 point higher, but holdout mIoU is 4.86 points lower, exposing session sensitivity.

Compact fusion removes 9.3% of parameters and 14.2% of GFLOPs; the full-width 640 sharing control removes 5.8% and 9.5%, respectively. On the Pi 5 (Table 5), fusion is $1.213\times$ faster at 448 and $1.218\times$ at 640,

reduces compute time by 18.57%/18.37%, and saves 21.87/34.41 MiB peak RSS. Thus 448 offers 16.4 FPS, whereas 640 trades speed for stronger detection at 7.3 FPS. No run reported thermal throttling.

5.2. Farm-Oriented Evaluation

Controlled length. For 150 physical shrimp (1,050 views), the largest clean detection at confidence 0.10 is retained when annotation IoU is at least 0.50. Report-GL3 calibration converts its box to length; estimates are averaged per shrimp.

Conditional MAPE spans 5.38–6.20%. Separate 448 has the lowest error and 46.10% valid-view coverage; fused 640 covers substantially more shrimp than separate 640 (90.67% versus 67.33%). Because rows contain different detected subsets, error and coverage must be read together. Paired common-view and unconditional diagnostics are reported in the supplementary material.

Historical archive. We sampled 258,809 common frames (71.9 h) at 1 FPS on three predeclared dates. Table 7 gives minute-balanced visible detections per frame and visible-feed periods from the fixed three-of-four-system consensus.

All systems reproduce the same daily ordering (Sep. 15 lowest, Sep. 30 highest). Fused/separate agreement precision–recall is .868/.859 at 448 and .895/.839 at 640; visible-count MAE is .243/.247 detections per frame. These are untracked cross-model agreement measures, not abundance accuracy. The 27 feed candidates contain 91 active minutes within 95 span minutes. Without event labels, they require visual confirmation and cannot rank systems or establish delivery, consumption, or clearance.

5.3. Ablations

Table 8 reports 448 validation controls. Stage 3 recovers $AP_{50:95}$ from .1947 after feed adaptation to .5606 while mIoU changes only .0003. Omitting Stage 2 retains detection but leaves feed IoU at .0236. The curriculum exceeds block alternating by 3.37 AP points and supervised-only Stage 2 by 2.68 points. The latter is the revalidated epoch-39 checkpoint after numerical

Table 4. Validation (V) and secondary session-disjoint holdout (H) performance. Compact fused rows are three-run means.

System	Input	Runs	AP _{50:95}		mIoU		Feed IoU	
			V	H	V	H	V	H
Separate S-0.75	448	1	.5713	.5834	.7215	.7591	.4620	.5449
Fused S-0.75	448	3	.5554	.5442	.7361	.7721	.4878	.5692
Separate S-0.75	640	1	.5920	.6032	.7509	.7684	.5184	.5614
Fused S-0.75	640	3	.5706	.5805	.7447	.7427	.5060	.5130
Separate full	640	1	.5920	.6032	.7664	.8283	.5471	.6743
Fused full	640	1	.5723	.5918	.7713	.7797	.5566	.5826

Table 5. Deployment complexity and Raspberry Pi 5 CPU results (means of three interleaved runs; four CPU threads).

Input	Execution	Params. (M)	GFLOPs	Compute (ms)	FPS	RSS (MiB)
448	Separate	6.207	8.761	69.820	13.525	178.214
448	Fused	5.628	7.513	56.856	16.401	156.344
640	Separate	6.207	17.879	162.090	6.015	216.026
640	Fused	5.628	15.332	132.311	7.328	181.620

termination at epoch 60. Secondary 640 feed-first and joint controls are supplementary.

Direct D routing gains 4.52 AP points for 22,920 parameters and 0.0285 GFLOPs (0.41% and 0.38% of the fused graph). The independently trained segmentation difference is not a D-only causal effect. Zero-shot D-input and BAG-gate sensitivities are therefore kept as supplementary diagnostics.

6. Discussion

Sharing PIDNet provides a practical CPU-edge gain rather than a cost-free accuracy improvement. Compact fusion removes 14.2% of the width-matched GFLOPs and raises Raspberry Pi 5 throughput by about $1.21\times$ at both resolutions, with lower peak memory. On the secondary session-disjoint holdout, however, compact AP_{50:95} is 3.92 points lower at 448 and 2.27 points lower at 640. Full-width fusion narrows the detection gap to 1.14 points, suggesting that width reduction explains part of the compact deficit. Segmentation is more session-sensitive: fused 448 improves feed IoU, whereas both 640 fused configurations trail their separate counterparts on the holdout despite competitive validation results. Farm deployment should therefore select 448 or 640 according to throughput and accuracy requirements, not assume one operating point dominates.

The ablations explain the task balance. Stage 2 feed adaptation damages detection, but Stage 3 restores AP while preserving the frozen segmentation state. Direct D features add 4.52 AP points at only 0.38% extra computation over P+I-only adapters. Together, protected adaptation, recovery, and P/I/D routing are more effective than detector-only or block-alternating controls for the reported balance; they are not claimed to maximize either task in isolation.

Application results must retain their operational definitions. Controlled length error is conditional on an eligible, annotation-matched detection, so coverage accompanies error. Historical counts are untracked visible-detection indices, not unique-animal abundance, and consensus feed periods are review candidates rather than feed mass, consumption, or verified events.

6.1. Limitations

The private partitions are session-disjoint but not farm-disjoint, and the holdout was accessed during legacy development; it is therefore a secondary, non-blind estimate of variation across capture sessions. Full-width and standalone controls have fewer repeats than compact fusion, and most ablations are single runs. Data restrictions also limit independent data-level reproduction.

The controlled-length IoU gate requires annotations and is unavailable in deployment; its intervals omit model-training and calibration uncertainty. The three-day archive has neither feed-event labels nor identities for tracking, so shared model errors may affect its consensus. Finally, the edge benchmark covers one Raspberry Pi 5, runtime, and mixed-FP16 configuration; energy, long-duration reliability, and other platforms remain unmeasured.

7. Conclusion

We presented a PIDNet-YOLO fusion that produces residual-feed segmentation and two-state shrimp detections from one PIDNet traversal. P/I/D adapters and a three-stage curriculum support disjoint feed-mask and shrimp-box datasets. The compact graph reduces width-matched computation by 14.2% and improves four-core Raspberry Pi 5 throughput by approximately

Table 6. Conditional controlled-length error with shrimp-level 95% bootstrap intervals (5,000 replicates).

System	Valid views	Shrimp covered	MAE, cm [95% CI]	MAPE, % [95% CI]
Separate 448	484/1,050 (46.10%)	141/150 (94.00%)	.503 [.452,.554]	5.38 [4.78,6.01]
Fused 448	417/1,050 (39.71%)	139/150 (92.67%)	.597 [.534,.664]	6.18 [5.50,6.95]
Separate 640	224/1,050 (21.33%)	101/150 (67.33%)	.562 [.495,.634]	6.20 [5.42,7.00]
Fused 640	428/1,050 (40.76%)	136/150 (90.67%)	.591 [.532,.652]	6.05 [5.44,6.68]

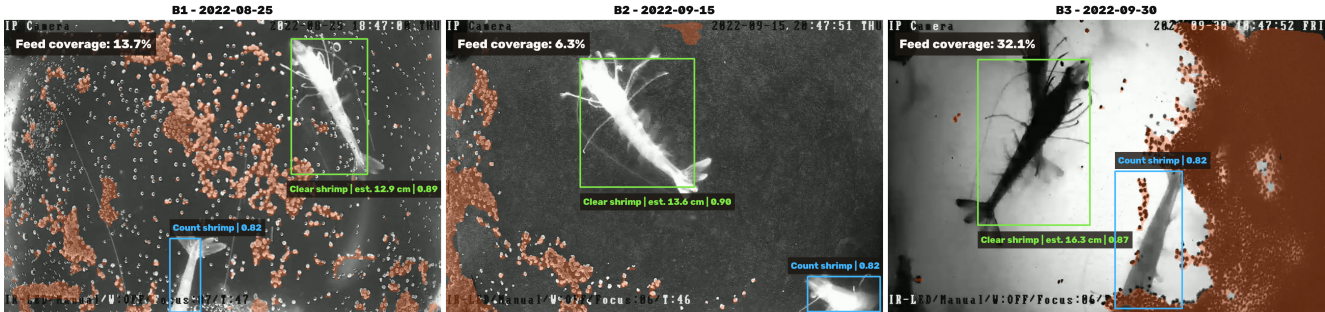


Figure 2. Representative observations from three historical dates at a private shrimp farm in Taiwan. Brown overlays show predicted feed pixels; boxes show clean/count detections, with Report-GL3 length estimates shown only for eligible clean detections. Feed coverage is visible image area, not feed mass. The examples are illustrative and have no frame-level feed-event ground truth.

Table 7. Historical observations; counts are separate/fused.

Day	Count 448	Count 640	Periods	Active/span (min)
Aug. 25	1.460/1.423	1.521/1.388	9	37/41
Sep. 15	1.013/0.998	1.038/0.966	9	31/31
Sep. 30	1.601/1.609	1.638/1.584	9	23/23

1.21 $\times$, with a measured accuracy–efficiency trade-off. Future work should test farm-disjoint transfer, tracked animals, labeled feed events, and long-term energy use.

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Table 8. Validation ablations at 448. D-interface rows are independently trained complete three-stage models; dashes denote unchanged or inapplicable deployment accounting.

Variant	AP _{50:95}	mIoU	Feed IoU	Params. (M)	GFLOPs
Training curriculum					
Post-Stage 2, no recovery	.1947	.7364	.4884	–	–
Three-stage BPKD-v3	.5606	.7361	.4878	–	–
No Stage 2	.5628	.4170	.0236	–	–
Block alternating	.5269	.7309	.4799	–	–
Supervised-only Stage 2	.5338	.7293	.4753	–	–
Detector interface					
P+I+D	.5606	.7361	.4878	5.628	7.513
P+I	.5154	.7550	.5249	5.605	7.484

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